



Islamic University of Technology
Department of Computer Science and Engineering

Lab 7

CSE 4404: Algorithms Lab
Summer 2023-24

Task A. Board Game

Time Limit: 1 second | Memory Limit: 512 MB

You are playing a board game where a token starts at position 0, and your goal is to reach position n .

On each move, you roll a standard 6-sided die and advance the token forward by the number of steps shown on the die. Thus, in one move, the token can move forward by 1 to 6 positions.

Determine how many different sequences of die rolls will move the token from position 0 to exactly position n .

Input Format

A single integer n the target position to reach. $1 \leq n \leq 10^6$

Output Format

Print the number of different sequences of dice rolls that sum up to n , modulo $10^9 + 7$. Since the answer can be large, print the result modulo $10^9 + 7$.

Examples

Sample Input	Sample Output
3	4
7	63
10	492

Task B. Book Shop

Time Limit: 1 second | Memory Limit: 512 MB

You are in a book shop which sells n different books. You know the price and number of pages of each book.

You have decided that the total price of your purchases will be at most x . What is the maximum number of pages you can buy? You can buy each book at most once.

Input Format

- The first input line contains two integers n and x : the number of books and the maximum total price. $1 \leq n \leq 1000$
- The next line contains n integers $h_1, h_2, \dots, h_n$: the price of each book. $1 \leq x \leq 10^5$
- The last line contains n integers $s_1, s_2, \dots, s_n$: the number of pages of each book. $1 \leq h_i, s_i \leq 1000$

Output Format

Print one integer: the maximum number of pages.

Examples

Sample Input	Sample Output
4 10 4 8 5 3 5 12 8 1	13

Marks Distribution

Task	Marks
Task A	50%
Task B	50%